

The Effect of Informal Economy on Budget Deficit in Egypt

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Abstract

The informal economy has attracted increasing international attention, as understanding its size and evolution is crucial for effective policymaking. Despite the growing global literature on methods of measuring the informal sector over the past three decades, research focusing specifically on Egypt remains limited. Therefore, this study seeks to estimate the size of Egypt's informal economy, thereby filling a gap in the literature and supporting policymakers in formulating appropriate public policies by illustrating the magnitude of informal activities.

Additionally, the study examines the impact of the informal economy on Egypt's public budget deficit. Using annual time series data for the period 1990–2020, the research employs the Autoregressive Distributed Lag (ARDL) model, the Bounds testing approach, and the Error Correction Model (ECM).

The findings reveal that the informal economy in Egypt represented, on average, about 63% of the country's Gross Domestic Product (GDP) during the study period. Moreover, the results indicate a positive relationship between the informal economy and the budget deficit, as informal activities and the associated tax evasion cause a significant loss of government revenues, thereby exacerbating the fiscal deficit.

The study concludes with several key recommendations, the most important of which is the need to address the underlying factors that drive individuals to participate in informal economic activities. This can be achieved by simplifying the process of obtaining business licenses and improving access to finance for small and medium-sized enterprises (SMEs).

Key Words: Informal Economy, Budget Deficit, ARDL, Egypt.

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I. Introduction

The primary objective of any government is to achieve key macroeconomic goals, particularly sustainable long-term economic growth. However, the rapid expansion of the informal economy in Egypt poses a significant challenge to economic stability policies, as it distorts several indicators that guide economic decision-making. Consequently, policymakers risk applying inappropriate policy measures due to inaccurate diagnoses of underlying economic problems. The informal economy leads to miscalculations in official statistics for key macroeconomic variables such as inflation, unemployment, and other indicators essential for effective policy formulation.

Economists generally agree that informal economic activities are not isolated from the formal economy; rather, they coexist and are closely intertwined. In many countries, the informal sector represents a substantial share of Gross Domestic Product (GDP). The purpose of this study is to evaluate the impact of the budget deficit on private investment, examine the relationship between both variables, and propose appropriate policy solutions based on the findings.

Previous research, including that of Schneider and Hassan (2015), has shown that informal economies have become too significant to ignore within the broader scope of economic activity, often exhibiting growth rates that surpass those of the formal sector.

The core problem addressed by this study lies in the coexistence of a persistent budget deficit and the growing size of the informal economy in Egypt. A substantial portion of the negative effects of the budget deficit could be mitigated by formalizing informal activities and integrating them into the official economy, thereby increasing tax revenues and strengthening public finances.

Accordingly, this study aims to estimate the size of the informal economy as a percentage of GDP and to measure its impact on Egypt's budget deficit over the study period.

Research Problem Statement

Despite continuous efforts by the Egyptian government to achieve fiscal discipline and sustainable economic growth, the persistence of a large informal economy remains a major obstacle to effective public finance management. The coexistence of a rising budget deficit and a significant informal sector reflects a structural imbalance in the economy.

The informal economy reduces the government's tax revenues through tax evasion and unregistered activities, which in turn widens the fiscal deficit and increases public debt. Moreover, the continuous fiscal deficit may further encourage individuals and businesses to operate informally to avoid higher taxes and declining public services. This creates a vicious cycle between the budget deficit and the informal economy.

Hence, the central problem of this research can be formulated as follows:

“To what extent does the expansion of the informal economy contribute to Egypt’s budget deficit, and how can integrating informal activities into the formal economy help improve fiscal sustainability?”

Or more formally:

The research problem lies in identifying and quantifying the impact of the informal economy on Egypt’s budget deficit, and in exploring the dynamic relationship between both variables to guide effective fiscal policy.

Research Objectives

The main objective of this study is to analyze the impact of the informal economy on Egypt’s budget deficit and provide policy recommendations to improve fiscal sustainability. Specifically, the study aims to:

1. Estimate the size of the informal economy as a percentage of Egypt’s GDP during the study period.
2. Examine the relationship between the informal economy and the budget deficit, including how informal activities affect tax revenues and fiscal stability.
3. Assess the impact of the budget deficit on private investment and economic growth in the context of a large informal sector.
4. Provide policy recommendations to formalize informal activities and strengthen public finance management in Egypt.

Research Hypotheses

Based on the research objectives and literature review, the following hypotheses can be formulated:

H1: The informal economy has a significant positive effect on Egypt's budget deficit.

H2: Higher budget deficits encourage greater participation in informal economic activities.

H3: The informal economy reduces tax revenues, thereby worsening fiscal imbalances.

H4: Formalizing informal activities and integrating them into the official economy can reduce the budget deficit and support sustainable economic growth.

These hypotheses allow for testing both the direct impact of informality on the fiscal deficit and the feedback effect from the deficit on informal activities.

Spatial and Temporal Scope of the Study

1. Spatial Scope: The study focuses on the Arab Republic of Egypt, examining informal economic activities and their impact on the national budget.
2. Temporal Scope: The study covers the period 2000–2023, which allows analysis of recent trends in the informal economy, public finance, and private investment.

This period also reflects economic reforms, fiscal policy changes, and the effects of global events such as the COVID-19 pandemic on Egypt's informal sector and public finances.

II. Literature Review

- **Theoretical Review**

Opinions about the phenomenon of the informal economy have varied according to the prevailing economic thought in the modern era. The spread of the informal economy in the world has shown us a reflection of each thought, and in this axis we will address the position of each of the following economic ideas:

First: the modernization perspective

There was a view that prevailed for a long time during the second half of the twentieth century that the informal economy was only a temporary phenomenon that would eventually disappear with the modernization of economies. In this view, the economic system is divided into two parts: (the formal economy and the informal economy). The formal economy is seen as a thriving phenomenon that symbolizes development, progress and modernity, while the informal economy is seen as a traditional activity that indicates backwardness and pre-modernity (Biles, 2008).

Several arguments can be drawn about the reasons for the growth of the informal economy from the perspective of modernization.

- 1- This perspective indicates that the informal economy will decline as countries enter a certain advanced stage of economic development. This points to a trend in which the informal economy is likely to experience a decline

corresponding to the increase in the country's GDP per capital (Schneider, 2012).

- 2- The expansion of informal economies results from the inability of the formal economy to create sufficient job opportunities for the urban labor force. In fact, it has been asserted that informal economies are more likely to grow when unemployment rates are high. Thus, the economic crisis is a catalyst for the growth of informal employment due to the decline of formal economies and the high number of unemployed in many developing countries that have experienced unprecedented population growth along with underdeveloped industrialization (Elgin, 2013).
- 3- The increase in the volume of migration from rural to urban areas will lead to the growth of informal economies, given the limited job opportunities in the formal economies. The factor of migration is well recognized in the literature but has rarely been pointed out in models of informal economies.

However, there is the fact that in recent years the informal economy has been flourishing all over the world and is becoming increasingly prevalent in both developing and developed countries. In addition, economic growth does not necessarily lead to a reduction in the size of the informal economy as expected from modernization perspective; In fact, the informal economy is

growing in some countries along with the development of modern economies (Heintz & Pollin, 2019).

Second: the new Marxist perspective

In the neo-Marxist perspective, the informal economy is seen as a by-product of contemporary capitalist restructuring, rather than as a consequence of the formal economy's inability to create adequate jobs. According to this perspective, the informalization of economies is a strategy used intentionally by capitalist firms to reduce costs, improve competitiveness, and weaken the power of unions. Globalization is seen as the primary driver for the expansion of the informal economy, as globalization leads to the growth of subcontracting activities, which lead to the expansion of informal employment. Thus, the informal economy is not separate from the modern economy, but rather linked to it. Thus, the informal economy is conceived of as a mode of production, and as a form of organization of work. (Sassen, 2019)

Neo-Marxist theory formulated some arguments for the reasons for the informal economy. . (Sassen, 2019)

- 1- the link between the formal and informal economies, it is likely that the growth of the informal economy is due to the expansion of industrialization and modernization, a trend that contradicts the assumptions of the modernization perspective. Thus, the relationship between informal economies and economic development in a

country is not a given, but needs to be justified and explained.

- 2- the informal economy is more likely to grow when economies become more globalized, as the globalization of production and trade has led to a boom in subcontracting and outsourcing activities. The growth of foreign direct investment in the country's economy may lead to the expansion of the informal economy. In general, in the light of the neo-Marxist perspective, the growth of the informal economy is seen as a result of the process of modernization and globalization.

Third: the neoliberal perspective

Scholars in the neoliberal perspective are interested in the effects of state regulation on economies and the responses of individuals to this. From this perspective, the informal economy is neither a by-product of contemporary capitalist restructuring nor an outgrowth of underdeveloped modern economies; Rather, it is the result of excessive state regulation, which incentivizes individuals to choose to engage in the informal economy in order to avoid onerous institutional costs. Thus the informal character can be conceived of as a representation of free market forces responding to the failure of state intervention. Researchers from this perspective highlight the ingenuity, entrepreneurship, and rationality of informal workers in creating income opportunities and alleviating their poverty in the context of the high cost of

entering the formal sector, as the neoliberal perspective showed that countries that have a high tax burden or many regulations imposed on their economies (for example (eg, time-consuming business registration, high-cost business organization) tend to have a larger share of the informal economy in their countries. (Gindling, & Newhouse, 2013).

- **Empirical Literature Review**

Spiro's (1997) study, the researcher studied the phenomenon of the growing size of the informal economy in Canada in the eighties and nineties of the twentieth century. He pointed out that there are intertwined relationships between the informal economy and the public budget deficit. As the informal economy weakens the tax revenue and then the fiscal deficit worsens, however, the increasing fiscal deficit pushes individuals to enter the informal economy; This is because of the growth of the public debt resulting from the continuation of the large deficit in the public budget and the subsequent payments of debt service creates a state of continuous increase in tax rates in light of the lack of provision of public services to compensate for this increase in tax rates. Consequently, individuals tend to engage in the informal economy and move away from the formal economy.

In the study (Savasan, 2003), the researcher estimated the size of the informal economy in Turkey during the period (1970-1998), using the Multiple Indicators and Multiple Reasons (MIMIC) method. The study indicates that the apparent effect of the growth of

the informal economy is a loss represented in the lost tax, and this loss in taxes resulting from the growth of the informal economy is referred to as the tax gap. The study indicates that Turkey has suffered from an increasing public budget deficit for decades, until the ratio of the public budget deficit to GDP reached 9%, and by combating the growth of the informal economy in Turkey, it was possible to control the public budget deficit.

Alkhdour (2011) study estimated the size of the informal economy in Jordan during the period (1976-2010), using the Multiple Indicators and Multiple Reasons (MIMIC) method. He concluded that the decrease in government tax revenues - due to tax evasion - leads to creating a gap between actual and potential tax revenues. Which in turn leads to an increase in the budget deficit, and this gap has various harmful consequences for the economy, and he indicated that the government may have to raise tax rates, and this is likely to push individuals to increase their participation in the informal economy to evade paying more taxes, which may lead This leads to an increase in the size of the informal economy, which ultimately exacerbates the distortions in the economy further and has significant distributional consequences.

Cicek & Elgin (2011), estimated the size of the informal economy in 78 selected countries during the period (1999-2007). They used cross-section regression models, as well as time-sector data (panel), as well as the OLS method. In all the estimated models, it was concluded that the size of the informal economy had a clear

and significant impact on the growing financial deficit in countries the sample.

IMF (2013) – Measuring the Informal Economy in the Caucasus and Central Asia ,This IMF study estimated the size of the informal economy in several Caucasus and Central Asian countries using the Multiple Indicators and Multiple Causes (MIMIC) model. The findings revealed that the informal economy represented about 26% of GDP in Kyrgyzstan and 35% in Armenia. The study also found that tax burden, labor market rigidity, and weak institutional quality were key factors associated with the expansion of the informal economy. This study is important because it applies a widely used quantitative model (MIMIC) in developing and transition economies.

The study of Enste (2015), referred to the size of the informal economy in a number of industrialized countries during the period (2003-2013), and indicated that the highest percentage was for Bulgaria's share of 33.2% of the GDP, and that the lowest percentage was for the United States US7.5%. Sweden and Portugal have average sizes in the sample, which are 15.8% and 20%, respectively. He pointed out that the best methods for estimating the informal economy are the indirect monetary methods and the multiple indicators and multiple causes (MIMIC) approaches. The study indicated that the increase in tax rates leads to an increase in the size of the informal economy; Which in turn leads to an increase in the budget deficit.

Özkök (2017) – The Analysis of Informal Economy with Monetary Ratio and Tax Audit Methods: The Case of Turkey This study estimated the informal economy in Turkey (1998–2015) using both the monetary ratio and tax audit methods. Results showed that the informal economy ranged between 25% and 50% of GDP, though with a gradual decline over time. The study is relevant as it applies alternative indirect methods beyond MIMIC and highlights the importance of combining multiple estimation techniques for accuracy.

Dell’Anno (2022) – Measuring the Informal Economy by a Structural Equation Modelling Approach Dell’Anno introduced the use of Structural Equation Modelling (SEM) to estimate the size of the informal economy, applying the approach to Italy.

Through Monte Carlo simulations, the study compared SEM to national accounts and monetary methods, showing that SEM can complement traditional estimation models. This study is important because it reflects methodological innovation in measuring the informal economy and offers a more flexible framework for empirical research.

Uygun & Gerçek (2023) – Measurement of Tax Gap and Analysis of Tax Gap Components in OECD Countries This research examined the tax gap—the difference between potential and actual tax revenues—in 35 OECD countries during 2005–2017, using panel data methods. It incorporated the informal economy rate, total tax rate, and GDP as explanatory variables and found that a larger

informal economy significantly increases the tax gap. This study is valuable for linking the informal economy with public finance imbalances and budget deficits through lost tax revenues.

Fiscal Sustainability and the Informal Economy: A Non-Linear Perspective (2024) This recent study analyzed the relationship between public debt, fiscal deficit, and the informal economy across 36 countries between 2004 and 2021.

Using a Panel Smooth Transition Regression (PSTR) model, it demonstrated that when the size of the informal economy exceeds a certain threshold, the relationship between public debt and the fiscal deficit becomes non-linear—with debt aggravating the deficit beyond that level. This is a valuable contribution as it explores dynamic and non-linear effects between the informal economy and fiscal outcomes.

Summary and Research Gap : While most empirical studies agree that the informal economy contributes to fiscal imbalances by eroding the tax base and weakening public revenue collection, few have quantitatively examined this relationship in Egypt using recent data and advanced econometric techniques.

Therefore, this study aims to estimate the size of Egypt's informal economy as a share of GDP and to empirically analyze its impact on the budget deficit, providing policy insights for integrating informal activities into the formal economy.

III. MEASURING THE INFORMAL ECONOMY IN EGYPT

The general ratio currency model (GCRM) is used to estimate the size of the informal economy in Egypt, the method can be mathematically represented through the following equation: (Feige, 1986)

$$Y_u = Y_o \frac{(C - rK * D)}{(rK + 1 * D)} \dots \dots \dots (1)$$

Where, C = Actual currency stock

D = Actual stock of checkable deposits

Y_o = Reported income

Y_u = Unreptred income

K = The ratio of currency to checkable deposit

rk = K_t - k(base)

To complete the measurement process, 2019 was chosen as a base year, in conjunction with the closeness of the ratio of money in circulation to demand deposits in that year according to (Feige, 1986,). As the growth rate of money in circulation decreased to demand deposits in that year, according to the data of the Central Bank of Egypt (Appendix A.1).

IV. Research Methodology

3.1 Model Specification

The autoregressive distributive lag (ARDL) model has been used to test cointegration or long run relationship between budget deficit and private investment and macroeconomic variables such as, inflation rate, Budget deficit and Household consumption expenditure.

This model is significant when it comes to model to variables of different integrated order. If F-statistics gives value that is greater than the lower critical values then can be conclude that there is cointegration between budget defecit and its determinants. but if F-statistic gives value less than the lower critical values it can be say that there is no cointegration. If we find cointegration then shall run error correction model (ECM), that the coefficient of CointEq (-1) should be negative and significant (Pesaran,2001).

The long-run relationship is measured according to the ARDL model, and the general formula for the long-run equation is as follows:

$$\Delta BD_t = \beta_0 + \sum_{i=1}^4 \beta_1 \Delta IBD_{t-i} + \sum_{i=0}^2 \beta_2 \Delta linformal_{t-i} + \sum_{i=0}^2 \beta_3 \Delta dlexch_{t-i} + \sum_{i=0}^2 \beta_4 \Delta lir_{t-i} + \sum_{i=0}^1 \beta_5 \Delta lrGDP_{t-i} + \phi_1 IBD_{t-1} + \phi_2 linformal_{t-1} + \phi_3 dlExch_{t-1} + \phi_4 lIR_{t-1} + \phi_5 lGDP_{t-1} + U_t \dots \dots \dots (2)$$

Where,

LBD = LOG of Budget deficit as a percentage of GDP

LINFORML = LOG of informal economy as a percentage of GDP

LEXCH = LOG of exchange rate (egyptian pound VS dollar)

LIR= LOG of lending interest rate

LRGDP= LOG of real GDP

β_0 = Constant

$\beta_1, \beta_2, \beta_3$ and β_4 are the coefficients

U_t = Error term

t = Time trend.

3.2 The Data

Data required for this study are Budget deficit as a dependant variable, and Informal economy as an independent variable, and control variables such as: exchange rate, lending interest rate, and real GDP. The estimation sample is from 1990 to 2020, for a total of 31 yearly observations. The data of exchange rate, lending interest rate, and real GDP were obtained from World Bank Data, while Budget deficit is sourced from The Egyptian Ministry of Finance, and informal economy is obtained from Author's calculations (Appendix A).

3.3 Descriptive Statistics

Table 1
Descriptive Statistics

variables	N	Mean	SD	Min	Max
LBD	31	0.73	0.33	-0.06	1.10
LINFORMAL	31	1.78	0.10	1.65	2.04
DLEXCH	31	0.04	0.08	-0.30	0.30
LIR	31	1.14	0.07	1.04	1.30
LRGDP	31	12.32	0.17	12.04	12.60

Source: Author's calculations, by Eviews

Descriptive statistics of sample data shows that the average value of Budet Deficit is 0.73%, and the informal economy has the average value of 1.78%, exchange rate has average value 0.04%, lending interest rate has average value 1.14%, and real GDP has average value 12.32%.

IV. Analysis and Discussion of Findings

Before carrying out the ARDL or Bounds test to cointegration, unit roots test was first conducted in order to examine the stationarity properties of the variables in the study.

• Unit Root Test

Table2
Unit Root Test

	Augmented Dickey Fuller (ADF)			Philipis-Perron		
variable	level			Level		
	constant	Constant&trend	None	constant	Constant&trend	None
LBD	(1.2138)	(1.8437)	(0.2735)	(1.4942)	(2.1120)	(0.4175)
Linformat	(1.8489)	(3.4226)*	(0.1324)	(2.0112)	(2.5057)	(0.1148)
DLexch	(3.8331)** *	(3.7007)**	(3.7433)** *	(4.8727)** *	(4.6175)***	(3.8536)** *
Lir	(3.1157)**	(4.1332)**	(0.9551)	(1.4725)	(1.5580)	(1.2752)
LrGDP	(1.5929)	(2.7330)	(3.1478)	(0.07539)	(2.1939)	(1.000)
	1 st Difference			1 st Difference		
	constant	Constant&trend	None	constant	Constant&trend	None
LBD	(4.0300)** *	(6.7434)***	(3.7348)** *	(4.0972)** *	(4.0647)**	(3.8259)** *
Linformat	(4.3680)** *	(4.2600)**	(4.4400)** *	(4.2773)** *	(4.1644)**	(4.3620)** *
DLexch	(6.2752)** *	(6.3366)***	(6.2457)** *	(6.3262)** *	(6.4259)***	(6.3230)** *
Lir	(3.2784)**	(3.1954)	(3.2239)** *	(0.7619)	(0.6607)	(0.8313)
LrGDP	(3.2725)**	(4.0010)**	(0.6388)	(3.2725)**	(3.2673)*	(0.6388)

Source: Author's calculations, by Eviews

Note: ***, **, *denotes rejection of the hypothesis at 1%, 5% and 10%

From the previous table, we note that some variables are stationary at their levels and others are stationary at the first differences, and therefore the ARDL methodology can be used to estimate the model.

- **Correlation matrix**

Table 3
Correlation matrix

	LBD	Linformal	Lexch	Lir	LrGDP
LBD	1.00000	0.267	0.213	-0.172	0.648
Linformal	0.267	1.00000	-0.119	0.061	0.629
DLexch	0.213	-0.119	1.00000	0.487	-0.207
Lir	-0.172	0.061	0.487	1.00000	-0.519
LrGDP	0.648	0.629	-0.207	-0.519	1.00000

Source: Author's calculations, by Eviews

According to the correlation matrix shown in Table 3, the independent variables have a weak correlation with each other. As a result, we can conclude that the model is free of multicollinearity problem.

- **Optimal lag length**

Table 4
Choosing the optimal lag length

Lag Length	LR	FPE	ALC	SC	HQ
0	NA	2.01e-11	-10.44160	-10.20163	-10.37025
1	221.2260	3.54e-15	-19.12432	-17.68450	-18.69619
2	38.16705	2.57e-15	-19.65791	-17.01824	-18.87300
3	30.76069	1.89e-15	-20.60249	-16.76297	-19.46080
4	40.01482*	8.62e-17*	-25.41977*	-20.38041*	-23.92130*

Source: Author's calculations, by Eviews

From the previous table, we note that there are four statistics confirming that the optimal slowdown period is one period: {Lag=4}, being significant for the largest number of differentiation criteria (FPE, AIC, HQ, LR).

- **ARDL Bounds Test**

Table 5
Cointegration Test

F-Statistic: 11.33	Bound Critical Value (no trend)	
Significant Level	I(0)	I(1)
10%	1.9	3.01
5%	2.26	3.48
2.5%	2.62	3.9
1%	3.07	4.44

Source: Author's calculations, by Eviews

As we refer to the table 5, the F-statistics (11.33) is more than the upper bound (4.44) at 1% significance level. therefore we can say that there is long-run cointegration (relationship) between the variables.

- **Coefficients Result**

Table 6
Long Run Coeffients Result

Lag Structure	(4.2.2.2.1)	
Dependent variable	BD	
Independent variable	Coefficient	(t_statistic)
Linformal	1.05	(2.032)*
DLexch	3.62	(2.920)**
Lir	-5.93	(6.223)***
LRGDP	0.42	(3.866)***

Source: Author's calculations, by Eviews.

Note: ***, **, *denotes rejection of the hypothesis at 1%, 5% and 10%

According to the result, we can conclude that the Informal economy has a positive effect on the budget deficit, as informal activities and the accompanying tax evasion lead to the state losing a large part of its revenues, and thus leading to an increase in the

deficit in the state's general budget. Exchange rate fluctuations (DEXCH): The exchange rate fluctuations reflect the effect of exchange rate shocks resulting from inflation and the increase in the volume of external debt, and it is clear that it has a positive impact on the budget deficit, Lending interest rate (IR): The interest rate in the model expresses the monetary policy followed, and it is clear that there is a negative effect of the interest rate on the general budget deficit. the future as a result of the high budget deficit resulting from the accumulation of debt, which leads them to increase current saving (Barro, 1974); The lower the interest rate, the less it will be an incentive for investment, and this will eventually lead to an increase in the budget deficit. and Real Gross Domestic Product (RGDP): The real GDP reflects the volume of public spending, and it is clear from the results that the real GDP has a positive impact on the budget deficit, due to the fact that the higher the volume of public spending, the higher the budget deficit.

- **Error Correction Model**

Table 7
Error Correction Model Result

variable	Coefficient	Std.Error	Prob.
CointEq(-1)*	-0.49	(8.69)***	0.0000

Source: Author's calculations, by Eviews.

According to the result, the Error Correction Coefficient (CointEq(-1)): It was significant, and had a negative sign. This corresponds to the conditions of applying the ARDL methodology to estimate the model.

- **The Quality and Suitability of the model used tests**

1- Breusch-Godfrey Serial Correlation LM Test

Table 8
LM Test Result

F-Statistic	0.1360	Prob. F(2,14)	0.7192
Obs*R-Squared	0.32991	Prob. Chi Square(1)	0.5657

Source: Author's calculations, by Eviews

In the Breusch-Godfrey Serial Correlation LM test for serial correlation, the P-value is 0.5657 which is greater than 0.05, so H_0 is not rejected at 5% significant level of confidence. Thus, there is no serial correlation in this model.

2- Heteroskedasticity Test: BPG

Table 9
Heteroskedasticity Test Result

F-Statistic	0.4996	Prob. F(1,26)	0.8408
Obs*R-Squared	4.9061	Prob. Chi Square(2)	0.7676

Source: Author's calculations, by Eviews

In the BPG test for heteroscedasticity test, the P-value is 0.7676 which is greater than 0.05, so H_0 is not rejected at 5% significant level of confidence. Thus, there is no heteroscedasticity in this regression model.

3- Ramsey RESET Test

Table 10
Ramsey RESET Test Result

	Value	Df	Probability
t-statistic	2.0911	11	0.0605
F-statistic	4.3729	(1,11)	0.0605

Source: Author's calculations, by Eviews

In the Ramsey RESET test for functional form and omitted variables, the P-value is 0.0605 which is greater than 0.05, so H_0 is

not rejected at 5% significant level of confidence. Thus, there is no misspecification in this regression model.

4- Structural stability test (CUSUM, CUSUMSQ)

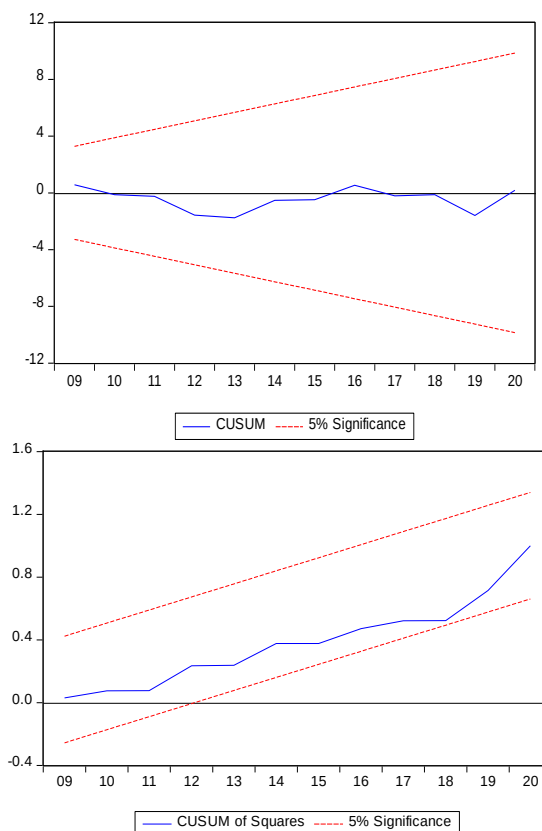


Figure 2: CUSUM and CUSUMSQ Test Result

By adopting CUSUM and CUSUMSQ test, we can view the stability test for long run model as shown in the graphs, confirming the stability in long and short run relationship between private investment and its determinants. (Brown et al, 1975).

V. Conclusion

This paper examines the relationship between informal economy and budget deficit in Egypt. The result found that informal economy has a positive and significant impact on budget deficit in the long run. Thus, Informal activities and the associated tax evasion lead to the loss of a large part of the state's revenues, and thus lead to an increase in the deficit in the state's general budget. The following policy implications have been developed based on the findings of the study. that the need to work on addressing the reasons that drive individuals to engage in the activities of the informal economy, by facilitating obtaining a permit to practice the activity, and providing the necessary financing for small and medium enterprises.

Results:

1- Impact of the Informal Economy on Budget Deficit:

- The informal economy has a positive and significant impact on Egypt's budget deficit.
- Informal activities and associated tax evasion lead to substantial revenue losses, which increase the deficit in the general government budget.

2- Impact of Exchange Rate Fluctuations

- Exchange rate fluctuations positively affect the budget deficit, reflecting the effects of inflation shocks and rising external debt.

3- Impact of Interest Rates

- Lending interest rates have a negative effect on the budget deficit. Higher interest rates reduce investment incentives, increase savings, and indirectly affect the deficit.

4- Impact of Real GDP

- Real GDP positively affects the budget deficit, as higher public spending associated with GDP growth increases the overall deficit.

5- Model Stability

- CUSUM and CUSUMSQ tests confirmed the stability of both short-run and long-run relationships between the budget deficit and its determinants.

6- Model Validity

- Tests for serial correlation (Breusch-Godfrey), heteroskedasticity (BPG), and functional form (Ramsey RESET) showed no econometric problems, confirming the robustness and reliability of the model.

Recommendations:

1- Formalization of Informal Activities

- Reduce the size of the informal economy by facilitating business registration and providing a clear legal framework for economic activities.

2- Support for Small and Medium Enterprises (SMEs)

- Provide financing and technical support for SMEs to encourage formalization and reduce informal sector participation.

3- Enhancement of Tax System Efficiency

- Improve tax collection mechanisms and broaden the tax base to minimize revenue losses caused by informal sector tax evasion.

4- Balanced Monetary and Fiscal Policies

- Implement policies to manage the impact of exchange rate fluctuations and interest rate changes on the budget deficit.

5- Institutional and Governance Improvements

- Strengthen institutional quality and transparency to monitor economic activities effectively and reduce distortions caused by the informal economy.

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Appendix A
Table A.1. Estimate the Informal economy

inf/ gdp %	Informal economy	V	GDP	r_k	k	D	C	year
54.88029	52728.98249	5.404736457	96080	1.034727	2.151392	5641	12136	1990
51.88272	58368.05542	5.983087805	112500	1.152286	2.26895	5752	13051	1991
56.43324	78498.64285	6.605565581	139100	0.978735	2.095399	6803	14255	1992
59.8647	92910.01682	6.337798105	155200	0.865313	1.981978	8212	16276	1993
57.07424	99879.92303	6.191621851	175000	0.956512	2.073176	9197	19067	1994
55.53415	113289.6759	6.448757666	204000	1.01077	2.127435	10115	21519	1995
57.12069	131043.4339	6.544243496	229415	0.954921	2.071585	11413	23643	1996
60.79815	161662.291	6.808870224	265900	0.836675	1.953339	13223	25829	1997
56.3609	161981.2345	6.59355786	287400	0.981275	2.097939	14070	29518	1998
57.47748	176800.7265	6.298374217	307600	0.942786	2.05945	15963	32875	1999
49.24011	167465.607	6.83783023	340100	1.267794	2.384458	14696	35042	2000
46.92577	168322.7319	6.711195929	358700	1.37964	2.496304	15287	38161	2001
48.55921	183990.8558	6.33559067	378900	1.299593	2.416257	17506	42299	2002
45.96471	191902.6761	6.211688389	417500	1.429395	2.546059	18954	48258	2003
45.31725	219924.5929	6.25338247	485300	1.464104	2.580769	21673	55933	2004
49.67657	267508.3334	6.004348553	538500	1.247869	2.364533	26656	63029	2005
55.76821	344480.2598	5.65276278	617700	1.002331	2.118995	35035	74239	2006
60.74385	452420.1605	5.67293345	744800	0.838317	1.954981	44430.1	86860	2007
75.91823	679847.7505	5.249757737	895500	0.470878	1.587542	65923.3	104656	2008
65.48155	682448.7393	5.695362067	1042200	0.705311	1.821975	64845	118146	2009
69.85652	842888.7212	5.637264063	1206600	0.598511	1.715176	78831	135209	2010
56.95433	780900.8404	5.512912785	1371100	0.960631	2.077295	80820	167887	2011
48.6751	815161.8712	6.1006885	1674700	1.294118	2.410782	80483	194027	2012
50.27238	935267.3228	5.406567858	1860400	1.221228	2.337893	103089	241011	2013
61.28097	1305284.724	5.188111673	2130000	0.822204	1.938868	139698	270856	2014
85.78608	2096525.987	4.89695731	2443900	0.301685	1.418349	206366	292699	2015
78.77553	2134344.119	4.728983218	2709400	0.417527	1.534191	226082	346853	2016
83.54911	2899154.259	4.905099749	3470000	0.336536	1.453201	288369	419058	2017
108.0962	4796662.82	5.40767804	4437400	0.033028	1.149692	381717	438857	2018
111.6664	5943222.192	5.76279665	5322300	0	1.116664	436329	487233	2019
98.48942	5766555.322	5.397596848	5855000	0.133791	1.250455	482010	602732	2020
63.36852								average

Source: Author's calculations, according to the Central Bank of Egypt data.

Table A.2. Bounds Test

Case 1: No Constant and No Trend				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LRGDP	0.425287	0.110004	3.866114	0.0022
LIR	-5.934828	0.953599	-6.223608	0.0000
DLEX	3.628627	1.242302	2.920889	0.0128
LINFO	1.055691	0.519487	2.032178	0.0649
EC = LBD - (0.4253*LRGDP -5.9348*LIR + 3.6286*DLEX + 1.0557*LINFO)				
F-Bounds Test				
Null Hypothesis: No levels relationship				
Test Statistic	Value	Signif.	I(0)	I(1)
Asymptotic: n=1000				
F-statistic	11.33698	10%	1.9	3.01
c	4	5%	2.26	3.48
		2.5%	2.62	3.9
		1%	3.07	4.44
Finite Sample: n=35				
Actual Sample Size	27	10%	-1	-1
		5%	-1	-1
		1%	-1	-1
Finite Sample: n=30				
		10%	-1	-1
		5%	-1	-1
		1%	-1	-1
-Bounds Test				
Null Hypothesis: No levels relationship				
Test Statistic	Value	Signif.	I(0)	I(1)
F-statistic	-5.525204	10%	-1.62	-3.26
		5%	-1.95	-3.6
		2.5%	-2.24	-3.89
		1%	-2.58	-4.23

Table A.3. Error Correction Model

ARDL Error Correction Regression				
Dependent Variable: D(LBD)				
Selected Model: ARDL(4, 1, 2, 2, 2)				
Case 1: No Constant and No Trend				
Date: 05/19/23 Time: 19:54				
Sample: 1990 2020				
Included observations: 27				
ECM Regression				
Case 1: No Constant and No Trend				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LBD(-1))	0.096887	0.109648	0.883617	0.3943
D(LBD(-2))	-0.052009	0.107363	-0.484423	0.6368
D(LBD(-3))	0.437979	0.108135	4.050296	0.0016
D(LRGDP)	5.896953	0.974334	6.052292	0.0001
D(LIR)	-1.726151	0.563560	-3.062942	0.0098
D(LIR(-1))	2.048347	0.670969	3.052821	0.0100
D(DLEX)	0.434440	0.320811	1.354196	0.2006
D(DLEX(-1))	-1.443573	0.303356	-4.758679	0.0005
D(LINFO)	-0.019558	0.322426	-0.060659	0.9526
D(LINFO(-1))	-1.033777	0.305300	-3.386103	0.0054
CointEq(-1)*	-0.492464	0.056646	-8.693668	0.0000

ملخص:

لقد جذب الاقتصاد غير الرسمي اهتمامًا دوليًا متزايدًا، حيث إن فهم حجمه وتطوره أمر بالغ الأهمية لوضع سياسات فعالة. على الرغم من تزايد الأدبيات العالمية حول طرق قياس القطاع غير الرسمي خلال العقود الثلاثة الماضية، إلا أن البحوث التي تركز تحديدًا على مصر ما زالت محدودة. ولذلك، تهدف هذه الدراسة إلى تقدير حجم الاقتصاد غير الرسمي في مصر، وبالتالي سد فجوة في الأدبيات ودعم صانعي السياسات في صياغة السياسات العامة المناسبة من خلال توضيح حجم الأنشطة غير الرسمية. بالإضافة إلى ذلك، تبحث الدراسة تأثير الاقتصاد غير الرسمي على عجز الموازنة العامة في مصر. باستخدام بيانات سنوية للسلسلة الزمنية للفترة ١٩٩٠-٢٠٢٠، تستخدم الدراسة نموذج الانحدار الذاتي الموزع مع التأخير (ARDL)، ومنهج اختبار الحدود، ونموذج تصحيح الأخطاء (ECM).

تكشف النتائج أن الاقتصاد غير الرسمي في مصر في المتوسط، حوالي ٦٣٪ من الناتج المحلي الإجمالي للبلاد خلال فترة الدراسة. تشير النتائج أيضًا إلى وجود علاقة إيجابية بين الاقتصاد غير الرسمي والعجز في الميزانية، حيث تتسبب الأنشطة غير الرسمية والتهرب الضريبي المرتبط بها في خسائر كبيرة لإيرادات الحكومة، مما يزيد من تفاقم العجز المالي. وتخلص الدراسة إلى عدة توصيات رئيسية، من أهمها الحاجة إلى معالجة العوامل الأساسية التي تدفع الأفراد للمشاركة في الأنشطة الاقتصادية غير الرسمية. ويمكن تحقيق ذلك من خلال تبسيط عملية الحصول على تراخيص الأعمال وتحسين الوصول إلى التمويل للشركات الصغيرة والمتوسطة الحجم (SMEs).

الكلمات المفتاحية: الاقتصاد غير الرسمي، العجز في الميزانية، ARDL، مصر.